

Improving Candidate Placement Success Rate

➤ Business Context & Introduction:

Placed is a fast-growing hiring agency that works with colleges and companies to improve campus recruitment outcomes. The company's core business objective is to match the right candidates with the right job opportunities, thereby improving recruiter satisfaction, reducing hiring inefficiencies, and increasing successful student placements. As Placed scales across multiple institutions and large candidate pools, one of the major challenges is that not every available student converts into a successful hire. Some students with strong academic records fail to get placed, while others with moderate academic performance but stronger practical skills, internships, or project portfolios succeed. This makes it important to identify which factors truly influence placement success.

The objective of this project is to perform exploratory data analysis on a student placement dataset containing 1,000,000 student records and 30 attributes. The analysis focuses on understanding academic, technical, behavioral, demographic, and experience-based factors that influence placement outcomes.

The main business question addressed in this report is:

What factors most strongly improve a student's likelihood of getting placed, and how can Placed use these insights to improve shortlisting and preparation strategy?

The analysis was performed using Python for data cleaning, EDA, correlation analysis, and segmentation, while Tableau was used to create interactive dashboards for stakeholder-friendly interpretation.

➤ Dataset Overview :

The dataset contains 1,000,000 student records and 30 columns. Each row represents one student profile, including academic performance, technical skills, soft skills, experience indicators, demographic information, and final placement status.

The dataset is large enough to support reliable analysis across multiple dimensions such as branch, college tier, skill level, experience, and candidate segments.

Category	Columns
➤ Academic	cgpa, backlogs, attendance, tenth_percentage, twelfth_percentage
➤ Institution	branch, college_tier, city_tier
➤ Technical Skills	python_skill, c++_skill, java_skill, ml_skill, web_dev_skill
➤ Soft Skills	communication_skill, teamwork, leadership, problem_solving, time_management
➤ Experience	internships, projects, github_projects, hackathons, certifications
➤ Assessment	aptitude_score, logical_reasoning, coding_contest_rating
➤ Demographics	age, gender, family_income
➤ Target Variable	placed

➤ Business Objective and North Star Metric :

The North Star Metric for this project is:

Placement Success Rate = Number of Students Placed / Total Number of Students

This metric directly reflects how effectively Placed converts its student candidate pool into successful hires. A higher placement success rate means better recruiter satisfaction, improved student outcomes, stronger institutional trust, and higher business growth for Placed.

From the dataset:

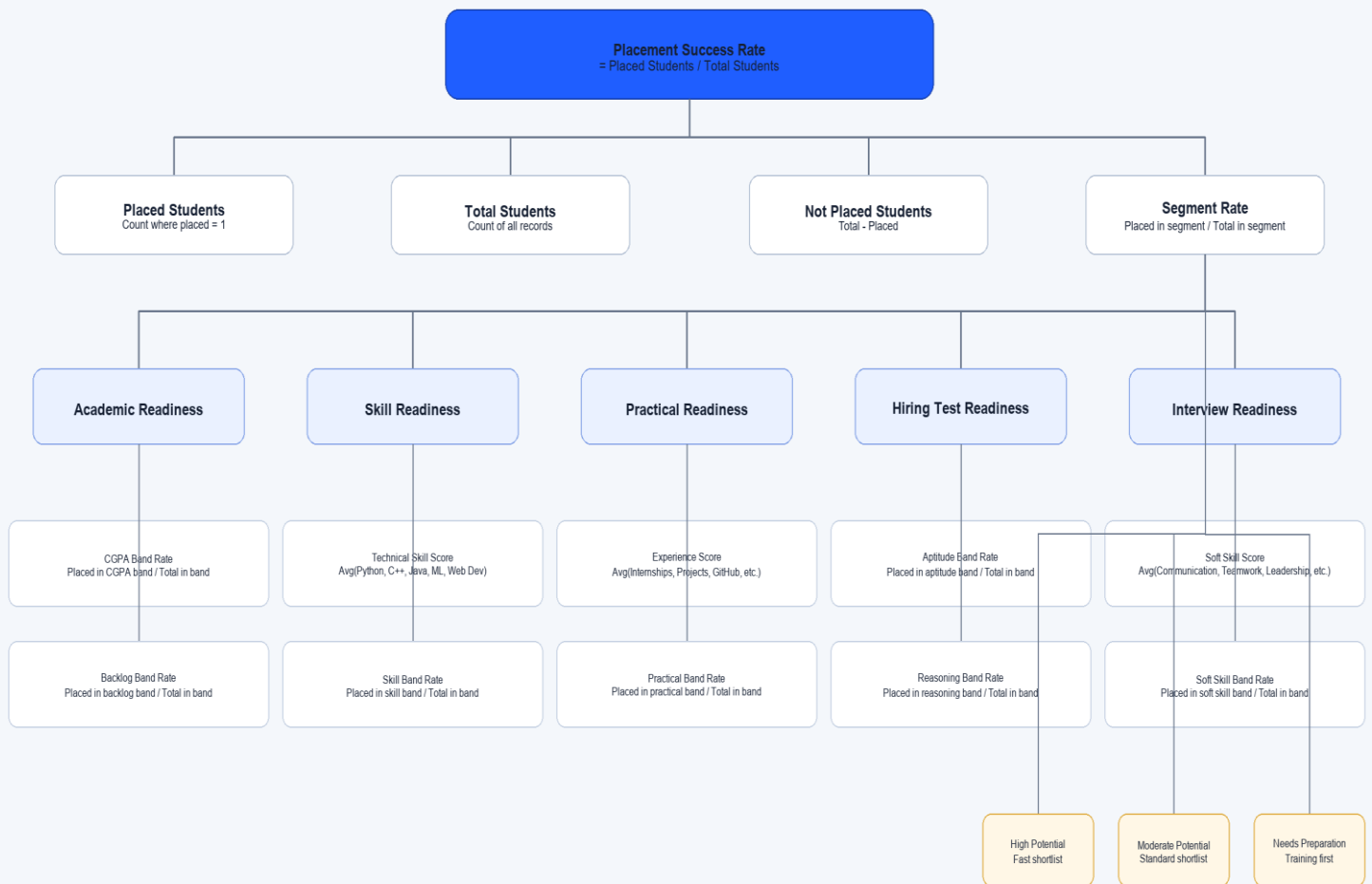
Metric	Value
Total Students	1,000,000
Placed Students	500,159
Not Placed Students	499,841
Placement Success Rate	50.02%

➤ Key Metrics & KPI Tree :

The North Star Metric is supported by several secondary metrics that explain why students are or are not getting placed.

The KPI tree helps connect business performance to measurable student attributes. It also helps Placed identify which factors are most useful for shortlisting and preparation planning.

KPI Tree: Candidate Placement Success Rate



Purpose: Break the North Star Metric into measurable drivers so Placed can identify which student attributes improve placement outcomes.

➤ Hypotheses :

	Hypothesis Statement	Findings
H1	Students with higher CGPA have a higher placement rate.	Students with CGPA ≥ 8 had a placement rate of 56.59%, while students with CGPA < 7 had 43.39%. Difference: +13.19 percentage points.
H2	Students with fewer backlogs have a higher placement rate.	Students with 0 backlogs had a placement rate of 52.25%, while students with 3 or more backlogs had 45.94%. Difference: +6.31 percentage points.
H3	Students with stronger technical skills have a higher placement rate.	Students with technical skill score ≥ 7 had a placement rate of 81.89%, while students with score < 5 had 40.12%. Difference: +41.77 percentage points.

H4	Students with higher aptitude scores have a higher placement rate.	Students with aptitude score ≥ 70 had a placement rate of 55.46%, while students with score < 55 had 45.53%. Difference: +9.92 percentage points.
H5	Students with better communication skills have a higher placement rate.	Students with communication skill ≥ 8 had a placement rate of 50.01%, while students with communication skill ≤ 4 had 50.10%. Difference: -0.09 percentage points.
H6	Students with more internships and projects have a higher placement rate.	Students with practical experience score ≥ 3 had a placement rate of 60.36%, while students with score < 1.5 had 39.12%. Difference: +21.24 percentage points.
H7	Placement rate differs by college tier.	Tier 1 colleges had 50.01%, Tier 2 had 49.97%, and Tier 3 had 50.07%. The difference between highest and lowest tier was only about 0.10 percentage points.
H8	Placement rate differs by branch.	The highest branch was ECE at 50.07%, and the lowest was CE at 49.94%. Difference: only 0.13 percentage points.
H9	Students with moderate CGPA but strong practical skills can still get placed.	Students with CGPA 7-8 and strong practical experience had a placement rate of 60.49%, while students with CGPA 7-8 and weak practical experience had 39.08%. Difference: +21.41 percentage points.
H10	Candidate segmentation helps identify high-potential students.	High Potential candidates had a placement rate of 90.28%, Moderate Potential had 71.25%, and Needs Preparation had 45.73%. This shows segmentation clearly separates placement likelihood.

➤ Data Cleaning & Preparation :

The dataset was inspected using Python. Initial checks included shape, column names, data types, missing values, duplicate records, and summary statistics.

The dataset shape was:

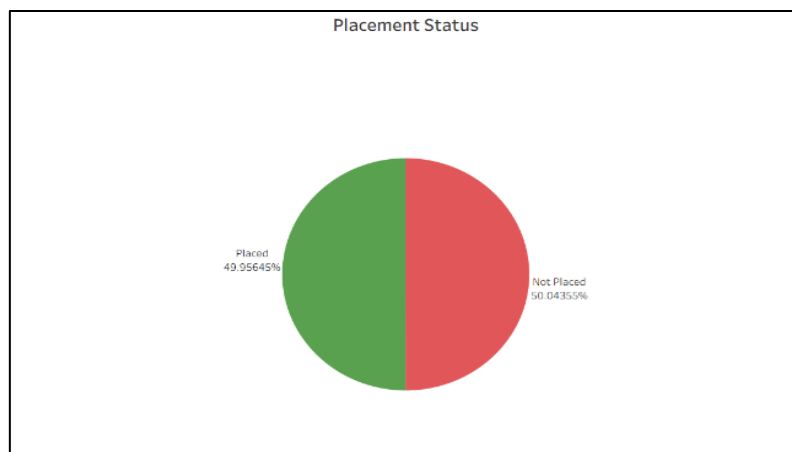
Rows – 10,00,000

Columns – 30

Data quality checks showed that there no missing values and duplicate values. The data set was already clean for analysis.

➤ Exploratory Data Analysis - Placement Overview :

The overall placement distribution is balanced.



The placement success rate is: **50.02%**

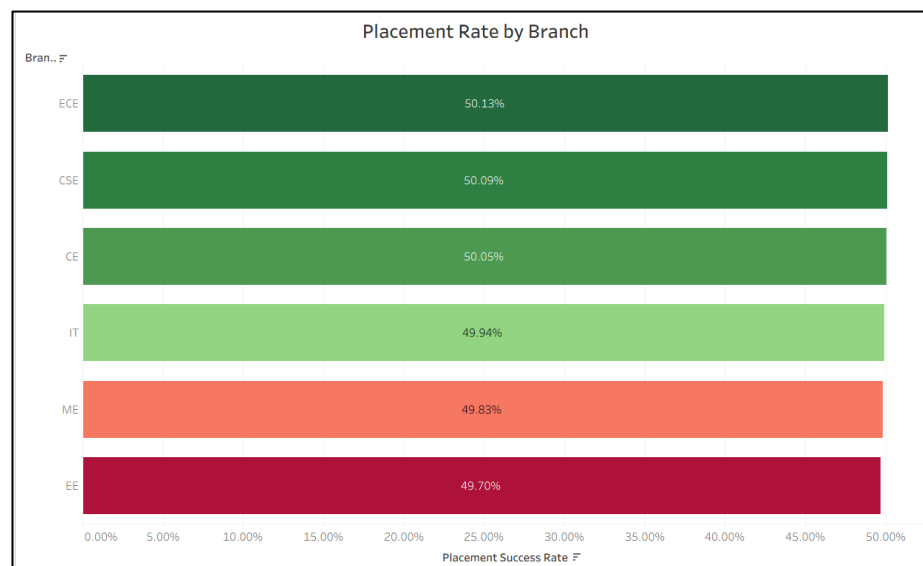
This shows that Placed is converting around half of the candidate pool into successful placements. Since the target variable is almost evenly distributed, the dataset is suitable for comparing placed and not placed students without major class imbalance concerns.

The current conversion level gives Placed a clear improvement opportunity. Even a small improvement in placement success rate can translate into thousands of additional placed students because of the large candidate base.

➤ EDA - Branch, College Tier, Gender, and City Tier:

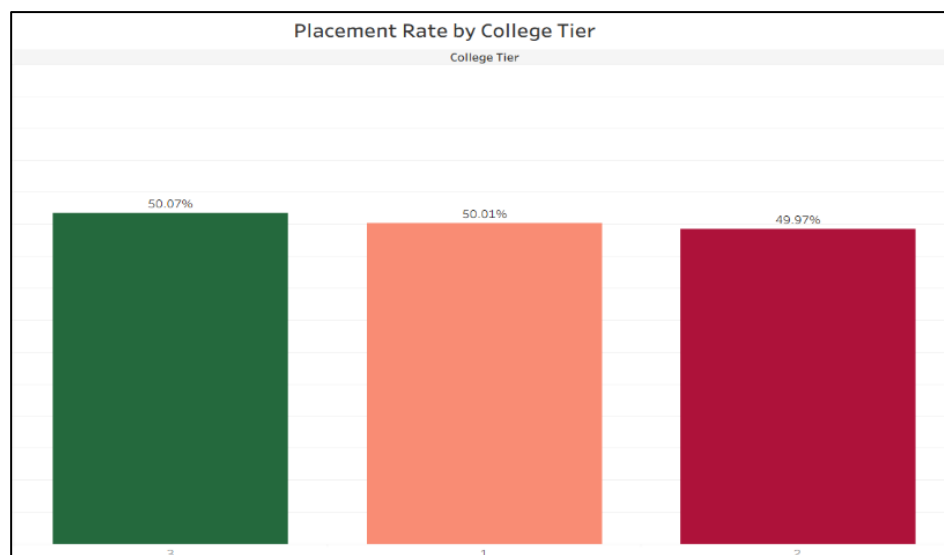
Placement rate was compared across categorical attributes such as branch, college tier, gender, and city tier.

1. Placement Rate by Branch:



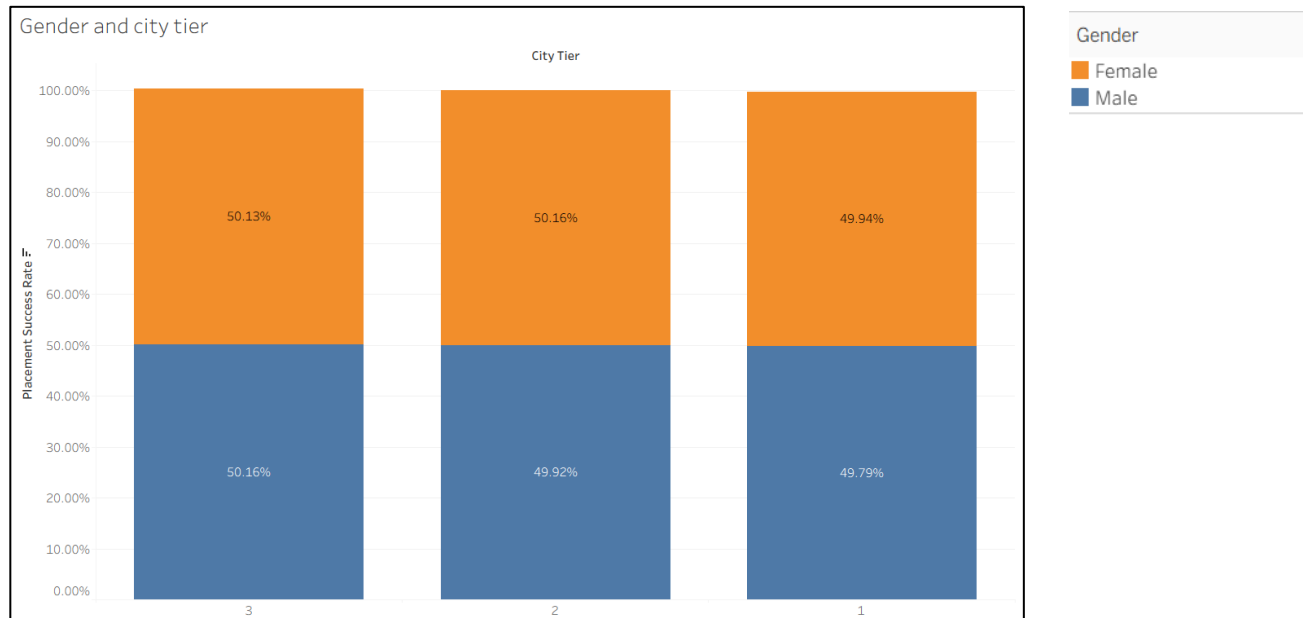
The difference across branches is very small. No branch shows a major advantage or advantage.

2. Placement Rate by College Tier :



College tier also shows almost no meaningful difference in placement success.

3. Placement Rate by Gender and City Tier:



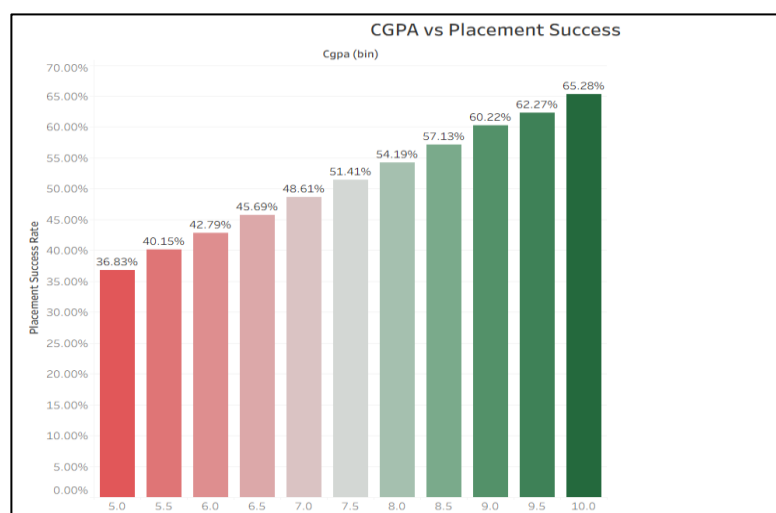
Findings:

Institutional and demographic factors show very small differences. Therefore, Placed should focus more on candidate-level readiness factors such as skills, CGPA, backlogs, internships, and project portfolios.

➤ EDA - Academic Factors :

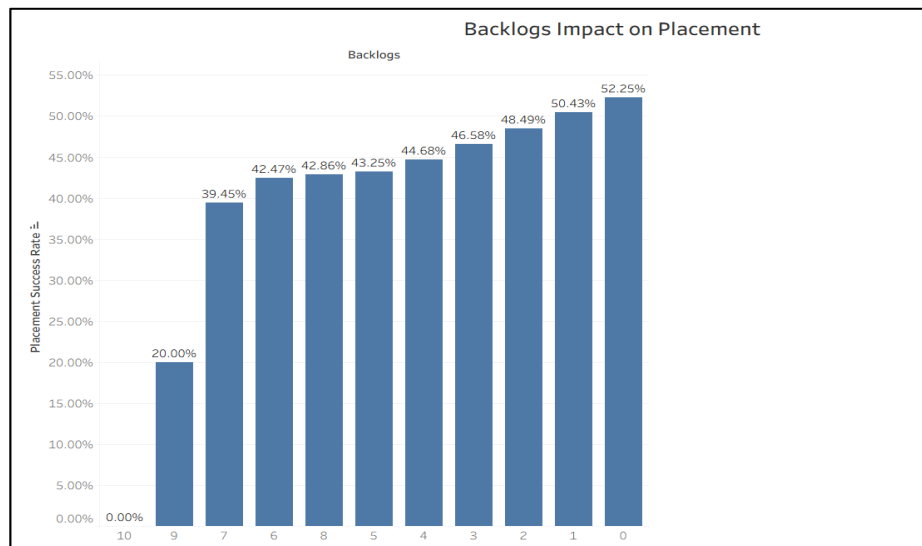
Academic performance was analysed using CGPA bands and backlog bands.

1. Placement Rate by CGPA Band:



CGPA shows a clear positive relationship with placement success. Students with CGPA above 9 have a placement rate of 61.22%, while students below 6 have only 39.12%.

2. Placement Rate by Backlog Band:



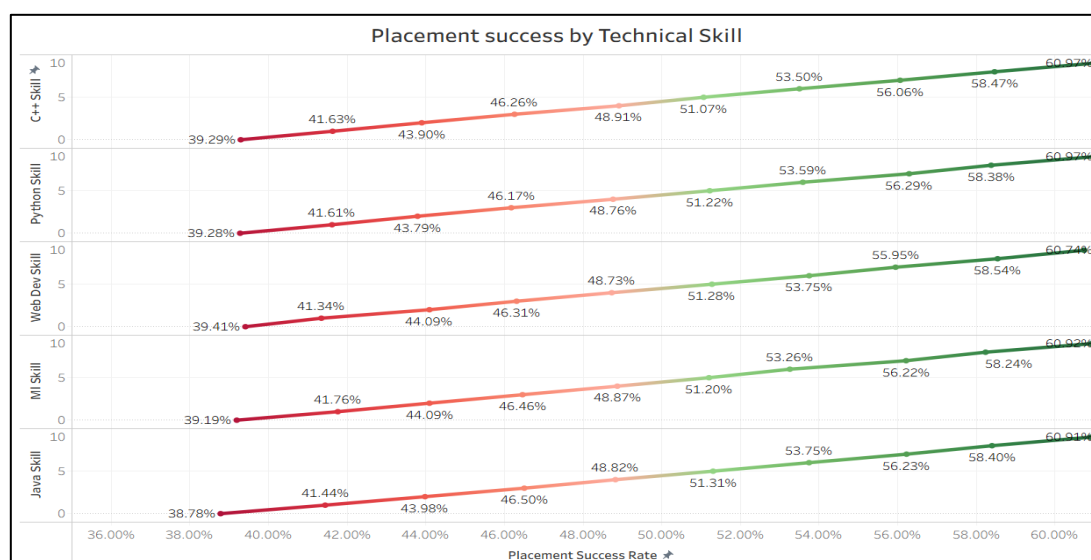
Backlogs show a negative relationship with placement. Students with no backlogs have a placement rate of 52.25%, while students with 3 or more backlogs have a lower rate of 45.94%.

Findings:

Academic strength matters, especially CGPA and backlog status. However, academics alone do not fully explain placement success, so skills and experience must also be evaluated.

➤ EDA - Skills and Assessment Factors:

Technical and assessment-related features were analyzed to understand their relationship with placement success.



Feature	Correlation with Placement
Technical Skill Score	0.310
Java Skill	0.140
Python Skill	0.139
C++ Skill	0.138
Web Development Skill	0.138
ML Skill	0.137
Soft Skill Score	0.111
Aptitude Score	0.086
Logical Reasoning	0.086

Technical skill score has the strongest relationship with placement among all engineered features. Individual technical skills also show consistent positive correlations.

➤ **Placement Rate by Aptitude Band:**

Aptitude Band	Placement Rate
85+	58.68%
70-85	54.70%
55-70	50.78%
40-55	46.62%
Below 40	42.20%

Aptitude also shows a clear positive pattern. Students with aptitude scores above 85 have a placement rate of 58.68%, compared with 42.20% for students below 40.

Findings:

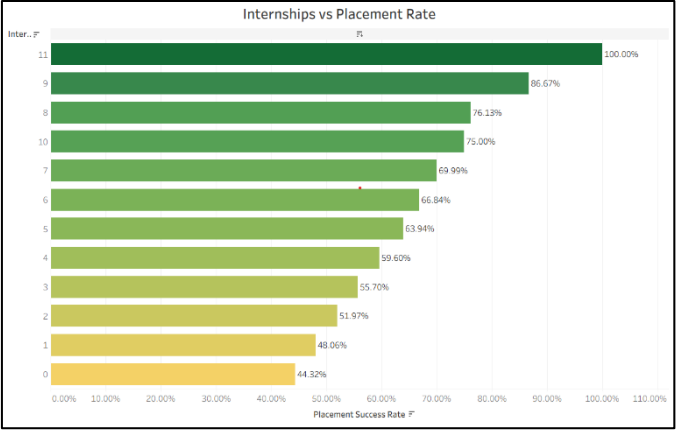
Technical readiness and aptitude performance are major indicators of placement success. These should be used as core shortlisting and preparation metrics.

➤ **EDA - Experience and Project Portfolio:**

Practical experience was analyzed using internships, projects, GitHub projects, hackathons, and certifications.

Experience-based features showed positive relationships with placement:

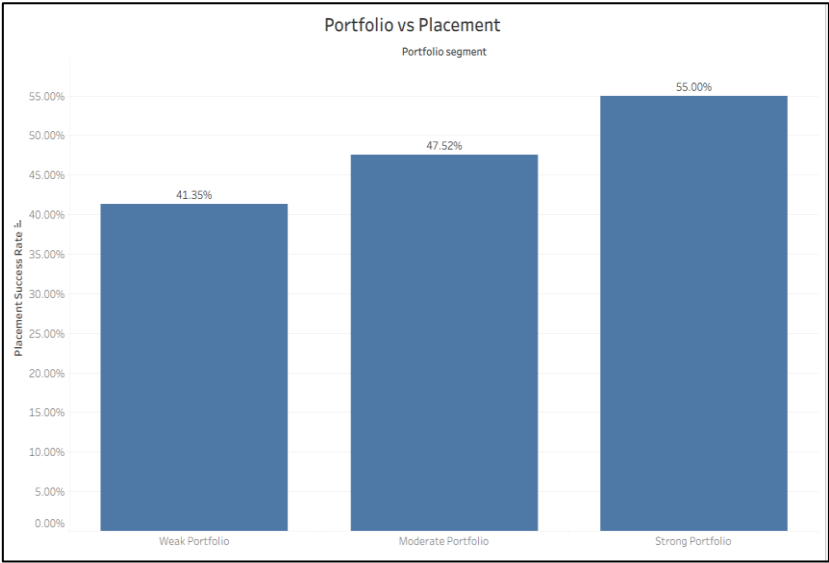
Feature	Correlation with Placement
Experience Score	0.136
Internships	0.094
GitHub Projects	0.078
Hackathons	0.075
Projects	0.066



Project portfolio strength was calculated using.

Portfolio Score = Projects + GitHub Projects

Students were grouped into weak, moderate, and strong portfolio segments.



This shows that students with stronger project portfolios convert better. The placement rate rises from 41.35% for weak portfolios to 55.00% for strong portfolios.

Findings:

Practical evidence of skills, especially projects and GitHub work, improves placement success. Placed should encourage students to build visible portfolios before recruitment drives.

- **Correlation & Feature Relationships :**
Correlation analysis was used to measure the strength and direction of relationships between numerical features and placement outcome.

Top positive relationships:

Feature	Correlation
Technical Skill Score	0.310
Experience Score	0.136
CGPA	0.115
Soft Skill Score	0.111
Internships	0.094
Aptitude Score	0.086
Logical Reasoning	0.086

Negative relationship:

Feature	Correlation
Backlogs	-0.041

- **Interpretation:**

The strongest placement driver is technical skill score, followed by experience score, CGPA, soft skill score, internships, aptitude, and logical reasoning. Backlogs have a negative relationship with placement, indicating that higher backlog count reduces placement probability.

Some features showed almost no relationship with placement, such as college tier, age, family income, attendance, 10th percentage, and 12th percentage. This suggests that recruiter conversion is more influenced by current employability readiness than by background attributes.

➤ **Tableau Dashboard Summary:**

Three main dashboards were developed in Tableau to communicate findings to stakeholders.

Dashboard 1: Executive Overview

Purpose: Show the overall placement picture and compare placement rates across branch, college tier, gender, and city tier.

Key message:

Placement success rate is 50.02%, and institutional categories such as college tier and branch show very small variation. This means Placed should not rely heavily on branch or college tier alone.

Dashboard 2: Placement Driver Analysis

Purpose: Identify the factors that influence placement success.

Key message:

CGPA, backlogs, technical skills, aptitude, internships, and project portfolios show stronger relationships with placement success than college tier or branch.

Dashboard 3: Candidate Segmentation

Purpose: Group candidates into actionable segments for recruiters and preparation teams.

Key message:

Candidate segmentation helps Placed prioritize high-potential students for recruiter matching and provide targeted preparation to lower-readiness students.

Interactive filters used in Tableau include branch, college tier, gender, city tier, placement status, candidate segment, portfolio segment, and backlog status.

➤ **Key Insights & Recommendations :**

- **Insight 1: Placement rate is balanced but has room for improvement.**

The overall placement success rate is 50.02%, meaning nearly half of the candidates remain unplaced.

Recommendation: Placed should use data-driven shortlisting and targeted training to improve conversion.

- **Insight 2: College tier and branch are weak differentiators.**

College tier placement rates range only from 49.97% to 50.07%, and branch rates range from 49.94% to 50.07%.

Recommendation: Avoid over-prioritizing college tier or branch. Focus more on student readiness indicators.

- **Insight 3: CGPA strongly supports placement success.**

Students with CGPA 9+ have a 61.22% placement rate, while students below 6 have only 39.12%.

Recommendation: Use CGPA bands to identify academic readiness, but combine them with skills and experience.

- **Insight 4: Backlogs reduce placement chances.**

Students with no backlogs have a 52.25% placement rate, while students with 3+ backlogs have 45.94%.

Recommendation: Create backlog recovery plans and separate preparation tracks for students with multiple backlogs.

- **Insight 5: Technical skills are the strongest driver.**

Technical skill score has the highest correlation with placement at 0.310.

Recommendation: Prioritize technical training in Python, Java, C++, ML, and web development.

- **Insight 6: Strong portfolios improve conversion.**

Strong portfolio students have a 55.00% placement rate compared with 41.35% for weak portfolio students.

Recommendation: Encourage project-based learning, GitHub portfolios, and demonstrable practical work.

➤ Conclusion:

This project analysed 1,000,000 student records to identify the key factors influencing placement success for Placed. The analysis combined Python-based EDA with Tableau dashboards for business storytelling.

The overall placement success rate was 50.02%, showing that Placed successfully converts about half of its candidate pool. However, the analysis revealed that placement success is not strongly influenced by college tier, branch, gender, or city tier. These attributes showed very small differences in placement rates.

The strongest placement drivers were technical skill score, experience score, CGPA, soft skill score, internships, aptitude, logical reasoning, and project portfolio strength.

Backlogs showed a negative relationship with placement success.

The analysis supports a clear business strategy:

1. Prioritize candidates with strong technical skills, strong portfolios, good CGPA, and internship exposure.
2. Provide targeted preparation for students with weak technical skills, low CGPA, limited project experience, or multiple backlogs.
3. Use candidate segmentation to support recruiter shortlisting.
4. Avoid relying only on institutional filters such as branch or college tier.
5. Build student preparation programs around technical skills, projects, internships, and aptitude improvement.

Placed can improve candidate placement success by shifting from broad demographic or institutional filtering to data-driven readiness-based shortlisting. The most effective strategy is to identify high-potential candidates early, prepare medium-potential candidates with focused support, and guide weaker candidates through skill-building, project development, and academic recovery before recruiter engagement.

.